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Final Project
RBIF 101

1) Biological background:

Introduction to the target: FtsA

Filamentous temperature-sensitive protein A (FtsA) is a potential target for novel broad-spectrum antibiotics. During bacterial cell division, the Z ring serves as an important scaffold for the proper recruitment and organization of a complex of key cytokinesis proteins, i.e. the divisome [1,2]. A bacterial tubulin homolog called FtsZ forms the core of this scaffold at the mid-plane of the cell [1,2]. Inhibiting the polymerization of FtsZ or its GTPase activity prevents successful division in most bacteria, and the former option is typically safer [3-5]. As such, this protein is already the target of a variety of promising antibacterial development projects, as well as a target for the repurposing compounds that are already FDA-approved [3-5]. While not quite as central to bacterial cell division, FtsA is also essential for proper Z ring formation and recruitment of divisome proteins [2]. FtsA is an actin-like, membrane-associated protein that binds to FtsZ monomers and anchors them to the inside of the membrane [2]. FtsA can also polymerize with itself, and its polymerization state is both coregulated with that of FtsZ and important early in the Z ring formation and divisome recruitment processes [2,6]. Due to the necessity of proper FtsA function for successful cytokinesis in many bacterial species, it seems that FtsA may represent a useful target for novel broad-spectrum antibiotics. Interrupting FtsA's ATPase activity, its interaction with other FtsA monomers, or its interaction with FtsZ all seem like potentially viable routes for disrupting proper cell division in a variety of bacteria [1,2]. To our knowledge, no major development has occurred around this target aside from some proof-of-concept work on peptide inhibitors of FtsA done by *Paradis-Bleau et al.* in 2005 [7]. While FtsA is often implicated as one of the more essential proteins for cytokinesis initiation, the specific proteins that are necessary for Z ring formation and divisome recruitment vary to some extent across bacterial species (aside from perhaps FtsZ) [1,2]. For example, under the right conditions *B. subtilis* can bypass FtsA for successful division [2]. With that major caveat in mind, here we evaluate the potential of FtsA as a viable and safe target for the development of a novel, broad-spectrum antibacterial agent.

Domain architecture and homology

FtsA is expressed in the common gram-negative and gram-positive bacterial species of clinical interest. **Table 1** summarizes relevant orthologous sequences for FtsA on UniProt that will be referenced throughout the report:

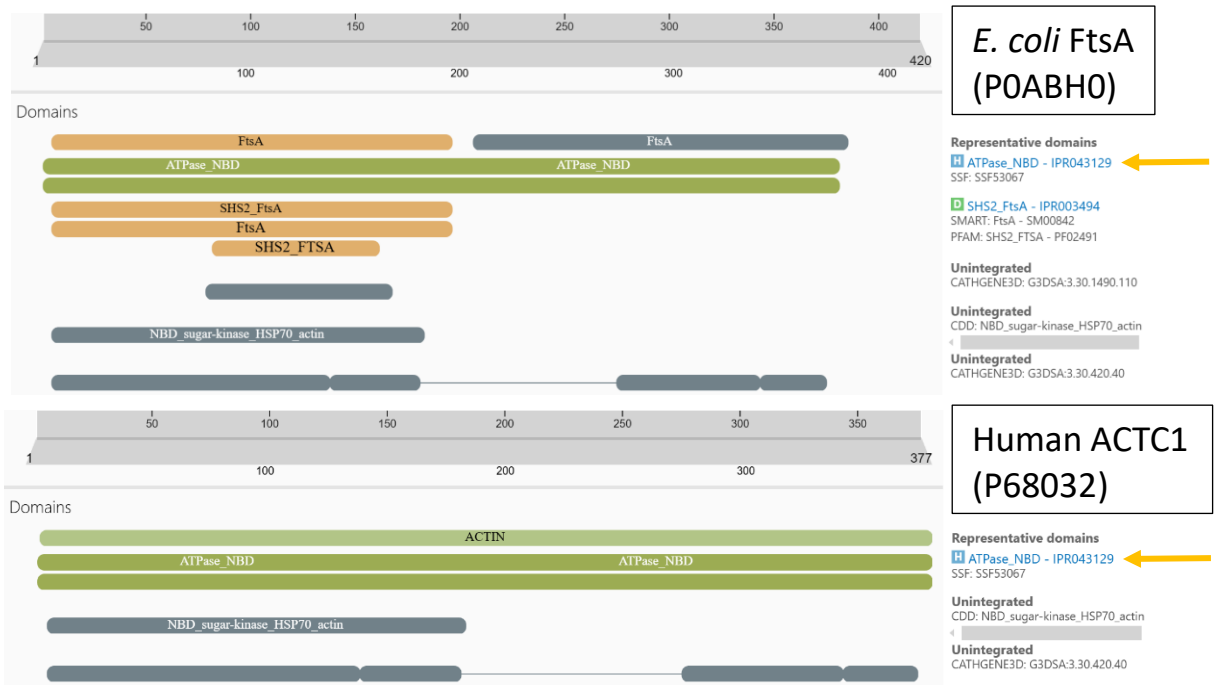
Table 1 – FtsA orthologs in species of interest (percent identities to sequences w/ PDB structures)

Species	Gram Status	Database	Reviewed?	Accession	% Identity to reference	
Escherichia coli (strain K12)	**negative**	**UniProt**	**Yes**	**P0ABH0**	**Reference**	**18.6**
Klebsiella pneumoniae	negative	UniProt	No	A0A377XQ65	98.1	18.7
Acinetobacter baumannii	negative	UniProt	No	A0A3S1BNK9	26.7	18.1
Pseudomonas aeruginosa (strain PAO1)	negative	UniProt	Yes	P47203	48.7	20.6
Staphylococcus aureus (strain MSSA476)	positive	UniProt	Yes	Q6GA27	18.6	98.9
Staphylococcus aureus (strain MRSA252)	**positive**	**UniProt**	**Yes**	**Q6GHQ0**	**18.6**	**Reference**
Streptococcus pneumoniae, serotype 2 (strain D39)	positive	UniProt	Yes	A0A0H2ZPT5	26.0	26.5
Bacillus subtilis (strain 168)	positive	UniProt	Yes	P28264	26.7	29.3
Enterococcus faecalis (strain V583)	positive	UniProt	Yes	O07111	28.1	25.0

As discussed earlier, FtsA is an actin-like protein that is necessary for successful cell division in bacteria [8]. It is composed of two different but structurally related "lobes", which are together part of a homologous superfamily of ATPase, nucleotide binding domains (IPR043129) [8,9]. One of these regions is considered to contain the ATP binding site, while the other typically contains a substrate binding site. That said, the ATP binding site is technically in a deep crevice located between the lobe domains [8,9]. This interesting structural feature of the ATP binding region, as well as its sequence, are well conserved within this homologous superfamily [9]. **Figure 1** summarizes the InterPro domain

architecture annotations for the FtsA sequence from *E. coli* and a distantly homologous human cardiac actin protein (ACTC1). While some of the other bacterial species from Table 1 have an additional FtsA C-terminal domain annotation (e.g. *E. faecalis* and *S. pneumoniae*), Figure 1 is fairly representative of the domain architecture common to the 9 species in Table 1 [10-18]. Commonalities between both the layout and content of the structural (CATH) domains and the sequence/profile-based ones can be seen in these distant homologs that share only ~12% identity:

Figure 1 - Bacterial FtsA domain architecture and homology to a human protein



It is through the domain homology highlighted above that we were able to find sequences for 69 distinct human homologs of the bacterial FtsA proteins (IPR043129) [9,10]. For brevity, we selected a subset of 10 homologs for further consideration that we considered representative of the potential risk for human harm. For the full list of human homologs, please see this list of InterPro results [10]. While a structural search using VAST or DALI may have unearthed a few more distant human homologs, we think this list likely represents the most pressing sources of risk. **Table 2** below shows the selected human homologs, along with their percent identities to the FtsA reference sequences from Table 1:

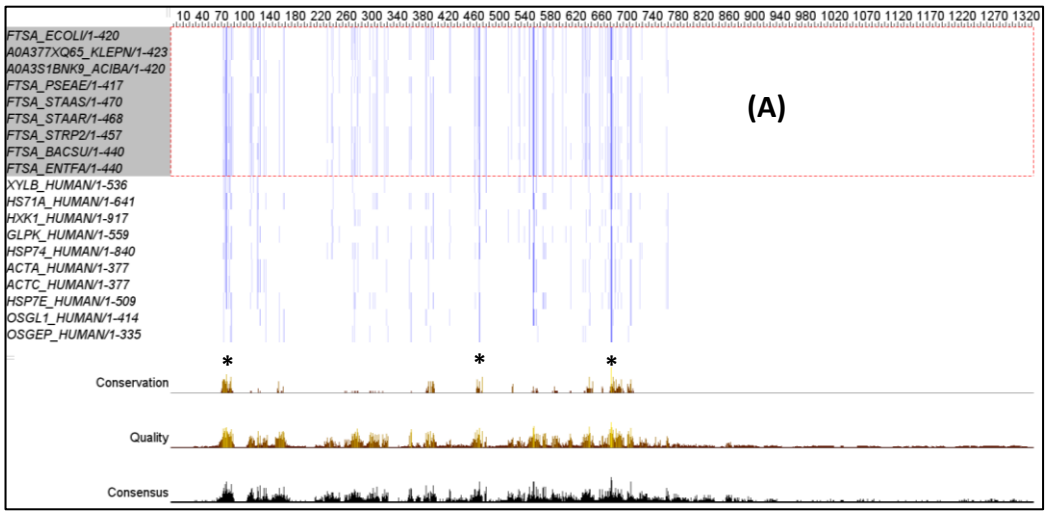
***Table 2** - Representative subset of the 69 human homologs of bacterial FtsA (found via InterPro homologous superfamily)*

Accession	Source Database	Name	Length	InterPro Family	%ID w/ K12 FtsA	%ID w/ MRSA252 FtsA
O75191	reviewed	Xylulose kinase	536	IPR043129	11.0	9.6
P0DMV8	reviewed	Heat shock 70 kDa protein 1A	641	IPR043129	11.7	13.0
P19367	reviewed	Hexokinase-1	917	IPR043129	13.6	12.2
P32189	reviewed	Glycerol kinase	559	IPR043129	11.2	10.0
P34932	reviewed	Heat shock 70 kDa protein 4	840	IPR043129	11.2	12.8
P62736	reviewed	Actin, aortic smooth muscle	377	IPR043129	12.2	11.4
P68032	reviewed	Actin, alpha cardiac muscle 1	377	IPR043129	11.9	11.4
Q0VDF9	reviewed	Heat shock 70 kDa protein 14	509	IPR043129	11.9	12.2
Q9H4B0	reviewed	tRNA N6-adenosine threonylcarbamoyltransferase, mitochondrial	414	IPR043129	11.4	10.1
Q9NPF4	reviewed	tRNA N6-adenosine threonylcarbamoyltransferase	335	IPR043129	12.5	11.6

Figure 2 at the top of the next page summarizes an MSA of the bacterial FtsA sequences from Table 1 and the homologous human sequences from Table 2 created using ClustalO. It can be seen in Figure 2A that there are only 3 or 4 areas where sequence is well conserved across most homologs. Figures 2A and 2B together show that sequence is, on average, better conserved (similarity or identity) within the bacterial FtsA orthologs than between these proteins and the human homologs (or even within human homologs). This is consistent with the percent identities to bacterial reference sequences shown in Tables 1 and 2. It is worth noting, however, that the percent identity between some FtsA sequences in Table 1 is still on the low/distant end of the spectrum (~18%). Encouragingly, the phylogenetic tree in Figure 2C based

Figure 2 – Overview of findings from MSA of sequences from Tables 1 & 2 (ClustalO)

***(A)** MSA in JalView with residues colored by percent identity and FtsA/bacterial sequences boxed in red. **(B)** Percent identity and similarity summary of matrices calculated by ClustalO & SIAS, respectively **(B)** Phylogenetic tree constructed via neighbor-joining by PAM250 in JalView. **(D)** E. coli FtsA structure (7Q6D) colored by percent identity in the MSA (as shown in (A)) [20].*



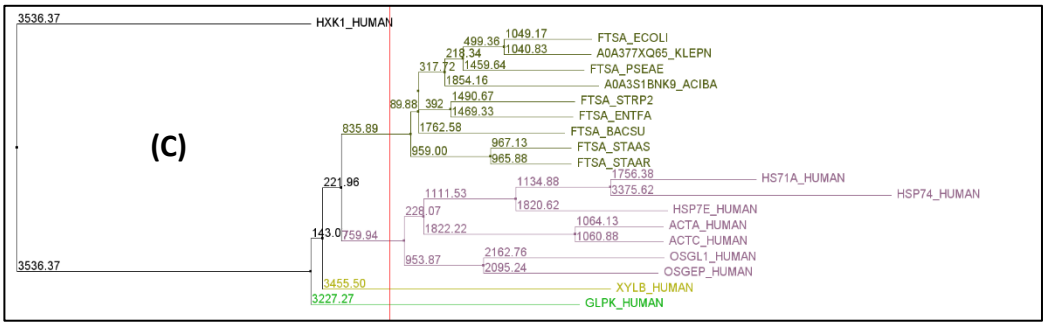
(B)

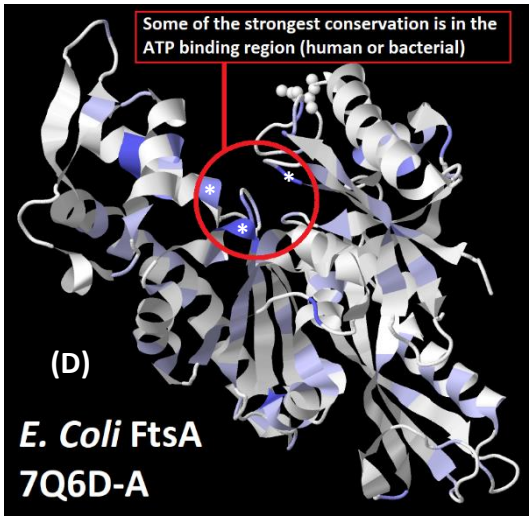
Percent identity overview (%ID matrix, ClustalO)

	Min	Max	Average
Bacteria-Bacteria	18.1	98.9	31.7
Bacteria-Human	9.6	14.3	11.8
Human-Human	10.0	98.4	14.6

Percent similarity overview (%SIM matrix, SIAS)

	Min	Max	Average
Bacteria-Bacteria	46.0	99.6	58.6
Bacteria-Human	15.6	32.4	24.4
Human-Human	10.0	99.7	26.0





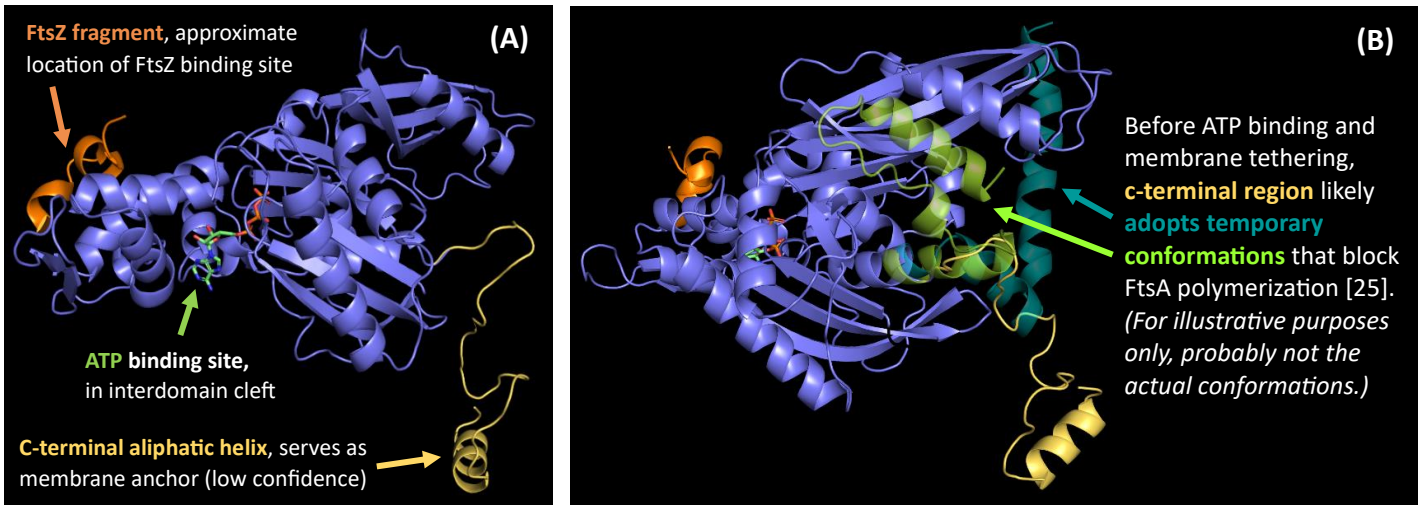
on PAM250 distance does accurately cluster the bacterial orthologs more closely to each other than to the distant human homologs selected from IPR043129. That said, the residues most well conserved across all species are somewhat unsurprisingly located in the ATP binding pocket, which is evident when mapping percentage identity from the MSA shown above onto a crystal structure of *E. coli* FtsA bound to ATP (see Figure 2D) [20]. This could pose a major challenge if the ATP pocket is the site targeted for inhibitor development, as inadvertently inhibiting one or more of the human homologs in Table 2 seems unacceptable for patient safety. Designing a sufficiently selective inhibitor around this pocket may be possible, but this homology certainly represents the greatest source of risk for FtsA as a viable target.

2) Comparative structural analysis:

E. coli and methicillin-resistant *S. aureus* (MRSA252) FtsA sequences from Table 1 have suitable structures on the PDB (7Q6D and 3WQT-A) [20-22]. To get a third bacterial structure, the sequence of the *P. aeruginosa* strain from Table 1 (P47203) was chosen as the target for comparative modeling, with the *E. coli* FtsA structure as the template. In Table 1 it can be seen that these two sequences had ~49% identity, which was for a pairwise alignment over 424 residues. This is within the safe zone for homology modeling according to *Krieger et al.* in their chapter from *Structural Bioinformatics* [23]. In Table 1, it is clear that the available structures from *E. coli* or *S. aureus* could have served as templates for two other species, although the sequence identity was so high it seemed any insights gained from the available PDB structures likely applied to those orthologs as well. Comparative modeling of the *P. aeruginosa* target was done using RosettaCM via Robetta (10 models sampled) [24]. The model with the lowest estimated errors throughout the "core" of the protein was chosen, and the vast majority of residue errors were < 1Å. **Figure 3** shows this homology model with key structural regions highlighted, which represent the three possible modes of disruption of FtsA activity mentioned in the biological background section. The flexible c-terminal regions in Figure 3 from RosettaCM are of low confidence, likely incorrect, and not included later in the analysis. That said, because this region is not present in the crystal structures, the

Figure 3 - Overview of FtsA structure and function via P. aeruginosa homology model from RosettaCM (based on 7Q6D)

***(A)** RosettaCM model of P. aeruginosa FtsA (indigo) with key functional areas highlighted. FtsZ fragment was mapped to this model via alignment with another bacterial structure (4A2A) [26]. **(B)** Low confidence c-terminal tail regions from RosettaCM provide an example of how it might adopt conformations that block FtsA-FtsA interactions before ATP binding and membrane tethering [25].*

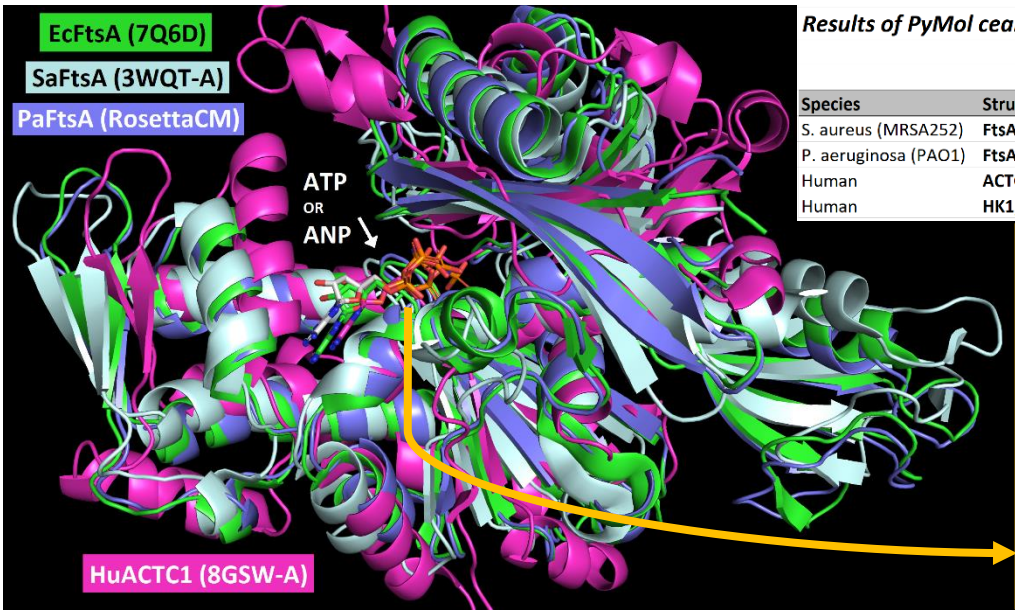


estimates from RosettaCM provide an opportunity to consider the flexibility of this important region and its propensity to form secondary structures that likely interact with the rest of the protein in functionally important ways [20,21,25].

Figure 4 below summarizes structural alignments conducted in PyMol against the *E. coli* FtsA structure. A structure for human cardiac actin (8GSW-A), the sequence of which was represented in Table 2 and Figure 1, appears in the alignments shown in Figures 4 and 5 [27]. This is a mutant structure chosen for its inclusion of AMP-PNP/ANP (very similar to ATP), however, the wildtype structure was extremely structurally similar (RMSD of 0.44 over most of the residues) [27,28]. It is clear from Figure 4 that the chosen human homolog has notable structural similarities with the FtsA proteins, however, the whole-structure alignments and corresponding RMSDs show that they are more similar to each other than to HuACTC1 overall. That said, in agreement with the findings in previous sections, the structural similarity between two human homologs and *E. coli* FtsA increases notably around the nucleic acid binding site (see RMSD values in reported Figure 4). The RMSD values do not drop for the FtsA structures when only considering this site, potentially indicating challenges around designing for safety/selectivity if an inhibitor were to bind there.

Figure 5 on the next page compares only the HuACTC1 and MRSA252 FtsA structures to highlight the domains, similarities, and differences. The coloring of the two largest lobes/domains for each protein was based on CATH annotation for the SaFtsA structure (3WQT-A), which only marks the first lobe [29]. It is worth noting that in these actin-like proteins, the lobe domains are often subdivided into two domains, an example of which can be seen here in the CATH annotation for a rabbit actin structure [30]. CATH's division of this type of protein into 4 domains can also be seen in the CATH annotation in Figure 1. That said, SCOP2 classifies this type of protein as a single conserved unit that is made up of two similar yet distinct sub-domains [31]. This can be seen by browsing the entries under the "Actin-like ATPases"

Figure 4 – Structural alignment and comparison of 3 bacterial orthologs (FtsA) and human homologs (e.g. ACTC1)



Results of PyMol cealign to E. coli (K12) FtsA structure (7Q6D)

		Whole structure		ATP binding region	
Species	**Structure**	**RMSD**	**Length**	**RMSD**	**Length**
S. aureus (MRSA252)	**FtsA**, 3WQT-A	3.95	352	3.89	32
P. aeruginosa (PAO1)	**FtsA**, RosettaCM (7Q6D)	1.03	368	2.60	32
Human	**ACTC1**, 8GSW-A	6.04	208	3.47	32
Human	**HK1**, 1DGK (not shown)	5.00	184	3.92	32

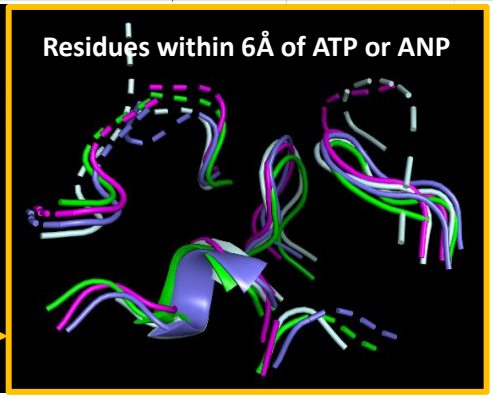
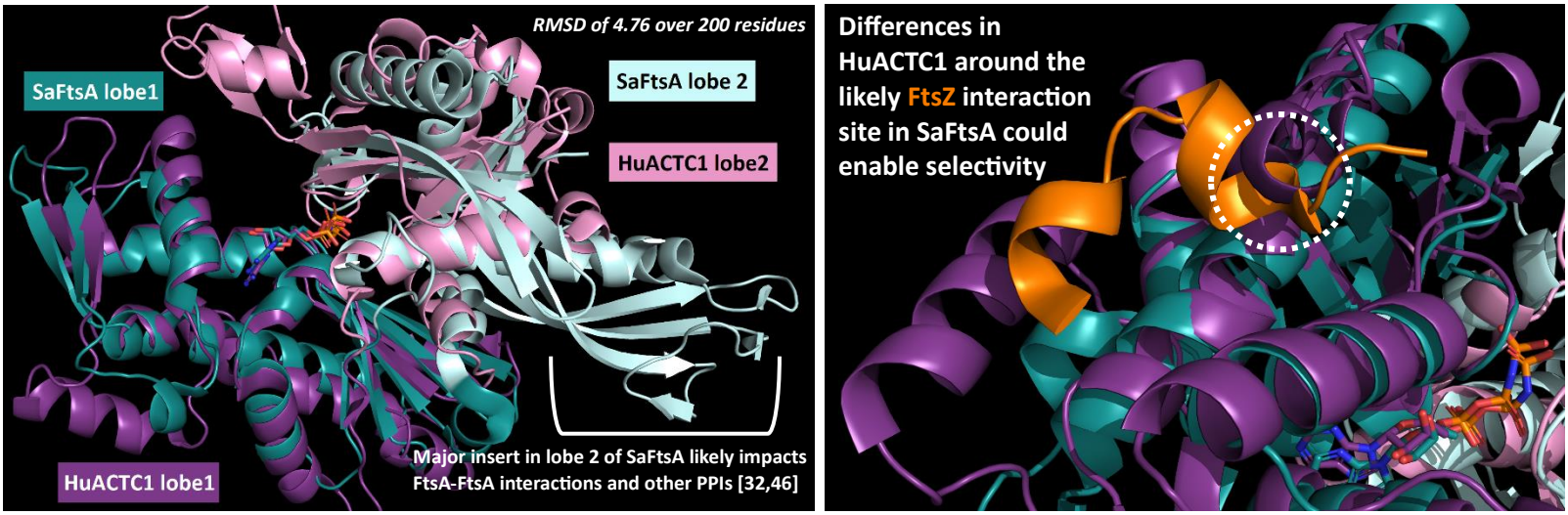


Figure 5 – Structural domains of human ACTC1 and MRSA252 FtsA compared, with key differences potentially relevant to small peptide or biologic agents noted



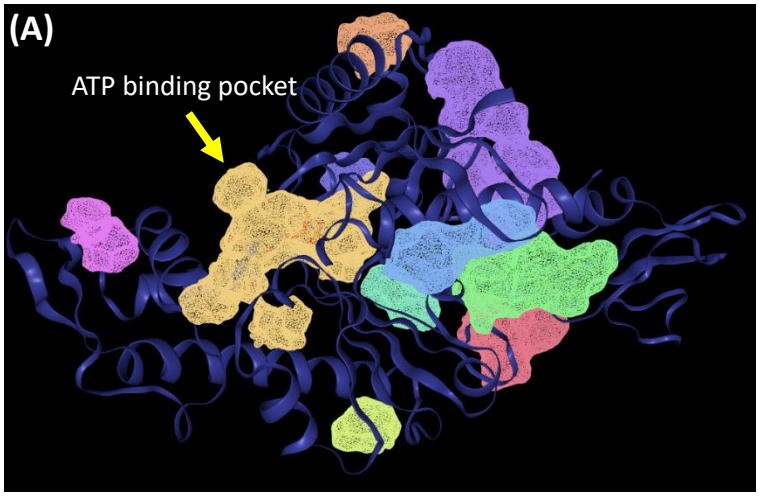
superfamily on SCOP2 (3000092) [31]. This is more in line with the InterPro/Pfam descriptions of this homologous superfamily discussed earlier in the biological background section. As such, we chose the two domain/lobe representation in Figure 5. The left panel shows that, overall, the bacterial and human homologs are fairly structurally similar and align relatively well given their low sequence identity (~11%). While distinct in the details, each lobe domain has a similar tertiary structure composed of both α and β secondary structures, featuring an architecture that CATH calls a "2-layer sandwich" [29-31]. The CATH topology associated with these structures is called "Nucleotidyltransferase; domain 5", while the SCOP2 fold is called a "Ribonuclease H-like motif" [30,31]. A structural feature of FtsA can be seen in the left panel of Figure 5 which is composed mostly of β-sheets "wrapped around" an α-helix. This does not exist in the same region of the structurally aligned HuACTC1 (or even HuHK1), but it is present in all of the FtsA structures considered (see Figure 4). The literature indicates that this region, called subdomain 1C, is an essential part of FtsA-FtsA polymerization, and it appears to likely perform an analogous role to that of the small domain on the opposite side of HuACTC1's lobe 2 above (called subdomain 1B) [25,32]. FtsA polymerization is important for Z ring formation and cell division, and subdomain 1C is distinct from its human actin counterpart in both size and position [32], so subdomain 1C of FtsA may provide targetable areas unique to bacteria [32]. As was done in Figure 3, in the right panel of Figure 5, an FtsZ fragment was mapped to SaFtsA via an alignment with a structure from *T. maritima* that captured an FtsA-FtsZ interaction (4A2A) [26,32]. Examination of this region in the aligned structures in PyMol revealed that HuACTC1 might be different enough from FtsA for a peptide (or other agent partially mimicking FtsZ binding) to selectively target this region and block FtsZ interaction. In a human hexokinase 1 (HK1) structure (1DGK), the region homologous to the FtsZ binding area shown above was *completely* blocked by an α-helix that connects two domains (not shown). Overall, a comparison of FtsA and homologous human structures revealed both major risks and potential for selectivity.

3) Binding site assessment:

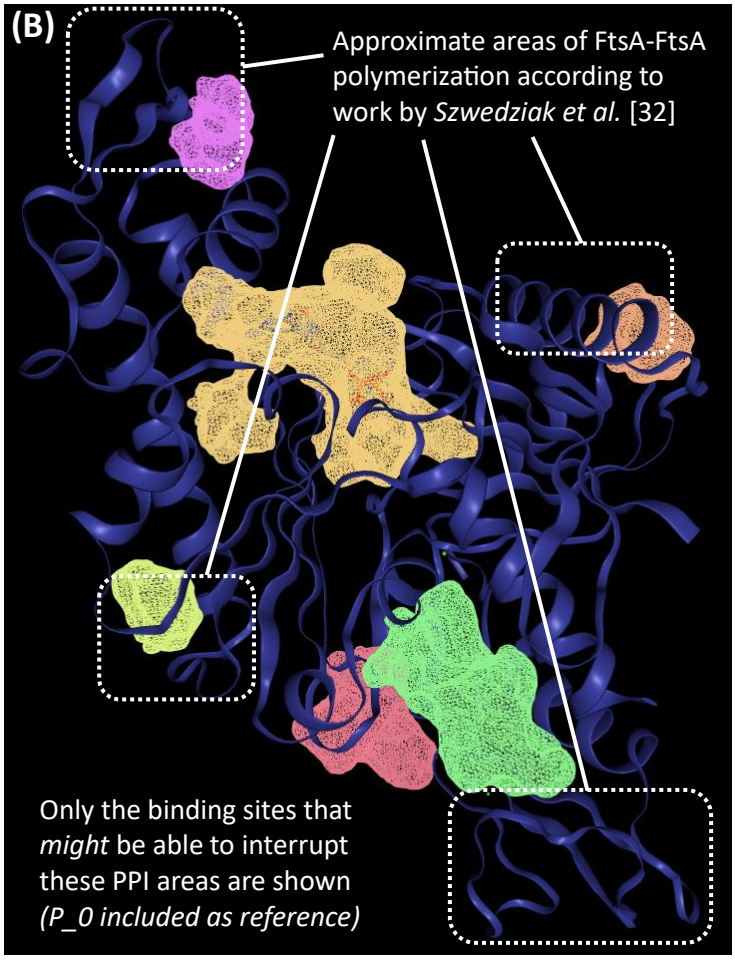
Figure 6 depicts the FtsA structure from the *S. aureus* strain MRSA252 considered earlier (3WQT), along with detected binding sites. These images were generated using the DoGSiteScorer tool from the Proteins.plus website, which could detect, rank, and illustrate the binding pockets of FtsA [33]. Chain A of 3WQT was used in the DoGSiteScorer analysis because FtsA is typically a monomer before it polymerizes into filaments [32]. The ATP binding site can be seen clearly in Figure 6, although in this structure it contains AMP-PNP, a chemical analog of ATP [34]. DoGSiteScorer identified several other pockets in this FtsA structure besides the ATP binding site. The table of binding site volumes and surface areas from the DoGSiteScorer analysis can be seen in Figure 6A, ordered by drug score. Somewhat unsurprisingly, the ATP binding pocket (P_0) was the largest of the ten detected binding pockets in 3WQT-A, with a volume of 1058.88 $Å^3$ and a surface area of 1107.59 $Å^2$. Binding site P_0 also had a drug score of 0.8 (the second highest) and a simple score of 0.6 (the highest, not shown). The only other site that was nearly as promising as the ATP binding pocket for draggability was P_1, which had a drug score of 0.86, a simple score of 0.5, a volume of 764.61 $Å^3$, and a surface area of 1040.12 $Å^2$. Figure 6B shows binding sites that may be in good positions for the direct disruption of protein-protein interactions necessary for FtsA polymerization, which will be discussed at the end of this section.

Figure 6 - Binding site analysis of MRSA252 FtsA structure (3WQT-A) using DogSiteScorer

***(A)** Comparable view of FtsA to that of Figure 5, with all pockets identified by DoGSiteScorer shown (top) along with their corresponding dimension metrics ordered by descending drug score (bottom). **(B)** Binding sites that may be well-positioned for direct interruption of FtsA-FtsA interactions necessary for polymerization (ATP pocket included as a reference) [32].*



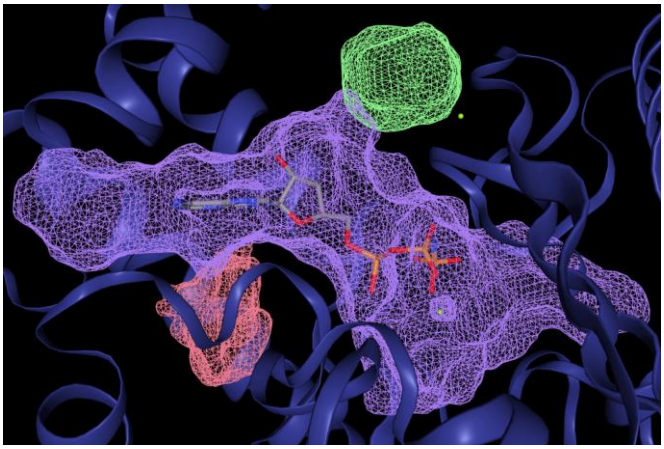
Color	Name	Volume Å³	Surface Å²	Drug Score
	P_1	764.61	1040.12	0.86
	P_0	1058.88	1107.59	0.8
	P_2	445.63	588.12	0.75
	P_4	305.92	401.97	0.66
	P_3	320.64	486.88	0.6
	P_6	141.89	411.4	0.44
	P_8	140.22	294.52	0.3
	P_5	155.01	265.48	0.26
	P_7	140.61	246.56	0.22
	P_9	108.16	219.16	0.21



DoGSiteScorer uses a geometric grid-based method involving the heavy atom coordinates to identify, cluster, and merge potential sub-pockets into the final binding sites [33]. It calculates a simple score for these sites, which takes the volume, enclosure, lipo surface, and hydrophobicity ratio into account [33,35]. "Lipo surface" refers to the lipophilic surface area of a binding pocket and gives an idea of the hydrophobicity of a site, which can be an important factor for ligand binding [33,36]. While the simple score can be used to evaluate druggability, the drug score is the output of a trained support vector machine (SVM) model that considers physicochemical factors determined to be the most effective for accurately classifying druggable pockets from undruggable ones [33], so it may be more reliable for assessing the druggability of a site. This is why only the drug score is shown in Figure 6A, although the simple scores are mentioned as needed. The drug score ranges from 0 to 1, with a higher score indicating a ligand pocket that is predicted to be more likely druggable [33]. As mentioned earlier, P_0 received a drug score of 0.8 and P_1 received a drug score of 0.86, both of which are in the "favorable" range and well above the average score of 0.51 for all the sites identified in Figure 6A [33]. The high druggability score of P_0 supports that the ATP binding site would probably be a suitable target for a competitive small molecule inhibitor. Unfortunately, despite its favorable properties, P_1 is not known to us to affect the biological function of FtsA in any way based on the background literature referenced so far [1,2,25,32,46]. As such, from a druggability point of view, the ATP binding pocket is by far the most promising one identified by DoGSiteScorer.

A sub-pocket within P_0 identified by DoGSiteScorer appears to be responsible for most of its potential druggability, as it was far larger and higher scoring than the others (see **Figure 7**). This sub-pocket (P_0_0) had a volume of 886.27 Å^3, a surface area of 953.05 Å^2, a drug score of 0.65, and a simple score of 0.8. It has 21 hydrogen bond donors, 77 acceptors, 44 opportunities for hydrophobic interactions, and a hydrophobicity ratio of 0.31. So, while P_0_0 is dominated by polar and charged residues, it is still an environment with opportunities for van der Waals interactions as well.

Figure 7 – Sub-pockets of the ATP binding pocket (P_0) identified by DoGSiteScorer in the MRSA252 FtsA structure (3WQT-A)



Name	Volume Å³	Surface Å²	Drug Score	Simple Score
P_0	1058.88	1107.59	0.8	0.6
P_0_0	886.27	953.05	0.65	0.8
P_0_1	99.78	229.82	0.01	0.0
P_0_2	72.83	54.52	0.5	0.0

Somewhat unsurprisingly, this sub-pocket is centered around where ATP binds and covers the majority of the interdomain cleft region discussed earlier in Sections 1 and 2. As such, P_0_0 likely represents a hot spot for ligand binding within P_0, the structural focal point of which is the 3 highly conserved GLY residues circled in Figure 2D earlier (darkest purple) and other nearby residues with strong structural conservation shown in Figure 4. The hot spot residues centrally located in P_0_0 and depicted earlier in Figure 4 for MRSA252 (3WQT) include ASP-10, ILE-11, GLY-12, SER-13, SER-14, SER-15, VAL-16, LYS-17, GLN-35, ILE-41, ASN-43, GLY-44, LYS-77, ASP-206, ILE-207, GLY-208, GLU-209, ASP-210, VAL-211, THR-212, GLN-213, ALA-231, GLY-232, ARG-233, THR-236, TYR-247, GLU-251, LYS-254, HIS-255, GLN-256, GLY-324, GLY-325, SER-326, ALA-327, ASN-328, LEU-329, PRO-357, GLU-358, and PHE-359. The homologous residues from the *E. coli* structure (7Q6D) are GLU-14, ILE-15, GLY-16, THR-17, ALA-18, LYS-19, VAL-20, SER-43, SER-84, ASP-210, ILE-211, GLY-212, GLY-213, GLY-214, THR-215, MET-216, ASP-217, ALA-235, GLY-236, ASN-237, THR-240, GLU-255, LYS-258, VAL-259, ARG-260, TYR-301, GLY-336, GLY-337, ALA-338, ALA-339, GLN-340, ILE-341, GLU-342, PRO-374, TYR-375, and THR-378. While not perfectly conserved, there is similarity/conservation of side-chain properties in these lists of residues, which is consistent with the structural conservation seen earlier. The same is true for the *P. aeruginosa* hot spot residues (not shown). This is to be expected based on the critical role of this region for this type of protein's function [37]. Consistent with the metrics reported earlier for P_0_0, there are some hydrophobic residues present in this central part of the pocket, but those lists are dominated by polar and charged residues.

There do not appear to be any biologically meaningful *and* sufficiently druggable sites outside of the ATP pocket in FtsA. First, none of the sites identified by DoGSiteScorer were near enough to the likely FtsZ binding region (mapped from 4A2A and seen in Figures 3 and 5 earlier) for direct disruption of interactions with FtsZ (not shown) [26,32]. P_1 looked promising as a druggable pocket, but it is not in a place that we know to be biologically useful. In Figure 6B, four regions are circled that indicate the approximate locations of FtsA-FtsA interaction sites that are important during polymerization, based on work by *Szwedziak et al.* investigating this phenomenon (see Figure 3 in that paper) [32]. If a small molecule of adequate size and shape could bind to any one of these four regions with high enough affinity, there would likely be potential for disrupting FtsA polymerization, and ultimately bacterial cell division [1,2,32]. While P_6, P_8, and P_5 are probably in the direct vicinity of three of the noted interface regions [33], it can be seen in Figure 6 that these pockets are quite small and had marginal to poor drug scores (and even worse simple scores of ~0). P_3 and P_2 have moderate to passable drug scores and decent volume and surface area [33], but these sites are likely a bit too far from their nearest FtsA-FtsA interface (bottom-right) for a small molecule to directly disrupt polymerization. There is a chance P_3 and P_2 are near key areas involved in FtsA's interaction with *other* divisome proteins [46], but we cannot not say this with confidence. So, where small molecule antibiotic agents are concerned, the ATP binding site appears to be the only viable target within FtsA.

4) Risk assessment and final recommendation:

The previous sections make it clear that, while FtsA is a promising target for novel broad-spectrum antibiotics due to its biological function [1,2,25,32], there is notable risk associated with it. The only confidently viable binding site for inhibiting FtsA activity (i.e. polymerization) using a small molecule agent appears to be the ATP binding pocket (P_0 from Section 3). Unfortunately, this pocket is also the focal point of FtsA's homology with a broad range of important human

proteins (see Figure 1, Table 2, Figure 2D, and Figure 4) [19]. The list of 69 human homologs represented by those shown in Table 2 includes various heat shock proteins, actins, actin-associated proteins, metabolic kinases, and even enzymes involved in preparing components of the translation machinery (among others) [19]. A compound that inhibits FtsA would need to be designed with selectivity as the top priority by carefully considering this long list of homologs. Accidentally inhibiting one or more of the targets in Table 2 could put a human patient in danger. For example, there are several hexokinases among the human homologs found [19]. These are important for metabolic activity and regulation, and reduced hexokinase activity can cause anemia [19, 38]. Glycerol kinase is another important metabolic protein in Table 2, the inhibition of which would likely be quite dangerous [39,40]. The unintended binding of a competitive inhibitor in the ATP pocket of human actin would interrupt its function, and cardiac actin was chosen for analysis and discussion to emphasize the potential risk associated with this type of homolog [41,42]. We were hoping that sufficiently promising small molecule binding sites would be located in biologically useful areas that were more structurally specific to the bacterial FtsA targets, thus granting some level of "baked-in" selectivity over the human homologs identified (see Figures 3, 5, and 6). Figure 6 shows that there were some sites identified with locations that were promising for specifically and directly blocking FtsA polymerization, but these are likely too small/unfavorable. As such, we are left with the ATP binding site (P_0), which is probably quite druggable but not specific to FtsA (see Figure 1, Table 2, Figure 2D, Figure 4, Figure 6, and Figure 7) [19].

It is worth noting that there is a precedent for designing successful inhibitors that compete with ATP binding in a target, and the ~30 kinase inhibitors that have been approved to date are a prime example of this [43]. That said, achieving adequate selectivity is a major challenge that must be overcome when designing these drugs [43]. It seems that designing a sufficiently selective inhibitor for the ATP binding pocket of FtsA would be quite challenging, as this region is responsible for some of the strongest conservation of sequence and structure seen between FtsA and its distant human homologs (see Figure 4 and the asterisk-marked regions in Figures 2A and 2D). It is worth noting that, based on the comparative structural analysis, and even parts of the binding site analysis, it may be possible to design a selective small peptide or other biological agent. While they are probably not targetable by small molecules, the regions of FtsA-FtsA or FtsA-FtsZ interaction discussed throughout this report may be feasibly targetable by larger ones. Some of these may be desirable spots to target on FtsA due to their biological relevance and specific presence on this protein. This may provide a way to bypass the challenges associated with selectively targeting the FtsA ATP pocket, but whether this benefit outweighs the added cost and complexity of developing a peptide or small biologic is outside of the scope of this report.

Our final recommendation to the company is that targeting FtsA for a novel broad-spectrum small-molecule antibiotic may not be worth the risk. The only viable pocket on this protein is the same location where the structural and sequential conservation with the identified human homologs is strongest. As discussed above, the list of human homologs is long and full of dangerous off-targets, so selectivity would need to be of the utmost priority for patient safety. Our outlook is that designing a compound with adequate affinity *and* specificity for FtsA in the clinically relevant species of bacteria, while maintaining sufficiently high selectivity over the many human homologs, may be prohibitively challenging. If FtsA were to be pursued despite our recommendation, we would suggest designing a competitive inhibitor around the parts of sub-pocket P_0_0 that are not directly central to ATP's binding, i.e. more toward the left or right sides of this pocket as shown in Figure 7. Structure and sequence conservation is likely to be somewhat lower a bit farther away from the exact "center" of the ATP binding site, opening up the potential for better selectivity over the human homologs while still physically occupying part of where ATP binds FtsA. That said, our final recommendation is that the potential of FtsA as a biologically promising target is likely not worth the risk due to the homology around the only plausible pocket. Larger modalities such as peptides or small biologics may provide viable ways to specifically target other parts of FtsA discussed throughout this report to achieve much stronger selectivity over the human homologs [7,44,45]. An evaluation of whether such an approach is feasible for this target is beyond the scope of this report.

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